

Dr. Erum Mehmood

Researcher | Assistant Professor

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OBJECTIVE

Dedicated and research-driven Computer Science academic with a Ph.D. and over a decade of tertiary-level teaching experience, seeking an associate professor position. Committed to delivering student-centered and innovative teaching in areas including Artificial Intelligence, Cybersecurity, Data Systems, and Software Engineering, while fostering critical thinking, practical problem-solving, and academic excellence through modern pedagogical approaches and technology-enhanced learning.

REASERCH AREAS

- AI-enabled WoT/IoT/VoT Security
- Big Data Analytics and Stream Processing
- Programming Languages Pedagogy
- Semi-Stream Join Algorithms
- Real-Time Data Warehousing
- NoSQL and Unstructured Data Systems
- Software Engineering and Software Architecture

RESEARCH PUBLICATIONS | 8 Impact Factor Publications | 2 Conf. | 380 Citations

Cyber Security and AI:

1. “Challenges in Detecting Security Threats in WoT: A Systematic Literature Review”, **Artificial Intelligence Review, Springer [Category: W, IF: 11.7, Rank: Q1]**, April 2025.
2. “Secure Image Encryption Using 5D Hyper-chaotic Keystream and 3D Pixel Scrambling with Adaptive Confusion-Diffusion” (Accepted by **7th International Conference on Advancements in Computational Sciences** at University of Lahore, Pakistan 10th-11th Feb 2026), Conference Proceedings published by IEEE.

Big Data Processing and Stream Architectures:

3. “DHSDJArch: an efficient design of distributed heterogeneous stream-disk join architecture”, **IEEE Access [Category: W, IF: 3.9, Rank: Q1]**, 2023.
4. “Distributed real-time ETL architecture for unstructured big data”, **Knowledge and Information Systems (Springer) [Category: W, IF: 2.7, Rank: Q2]**, 2022.
5. "An efficient data access approach with queue and stack in optimized hybrid join", **IEEE Access [Category: W, IF: 3.9, Rank: Q1]**, 2022.
6. “Challenges and Solutions for Processing Real-Time Big Data Stream: A Systematic Literature Review”, **IEEE Access [Category: W, IF: 3.9, Rank: Q1]**, 2020.
7. “Optimizing Semi-Stream CACHEJOIN for Near-Real-Time Data Warehousing” published in **Journal of Database Management (JDM) [Category: X, IF: 1.139, Rank: Q2]**, 2020.
8. “Performance Analysis on NotOnly SQL streams join using MongoDB for Real-time Data warehousing”, published in **IEEE Access [Category W, IF: 3.9, Rank: Q1]**, 2019.
9. “Optimization of Cache-Based Semi-Stream Joins (Accepted by **IEEE International Conference on Cloud Computing and Big Data Analysis** at Chengdu, China 28th -30th April 2017) Conference Proceedings published by IEEE..

Pedagogical Innovation in Programming Education:

10. “Curriculum, Teaching and Learning, and Assessments for Introductory Programming Course”, **IEEE Access** [Category: W, IF: 3.9, Rank: Q1], 2020.

EDUCATION

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- | | | |
|---|------------------|------------------------------------|
| • Ph.D. Computer Science
CGPA (3.95/4.00)
<i>Doctoral Thesis: “Integrated Real-Time Distributed Stream-Disk Processing Architecture for Unstructured Big Data”</i> | 2018-2024 | UMT Lahore, Pakistan |
| • M.Phil. Computer Science
CGPA (3.9/4.00) | 2014-2017 | NCBA&E Lahore, Pakistan |
| • MSc. Computer Science
CGPA (3.25/4.00) | 1999-2002 | BZU Multan, Pakistan |

SKILLS

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- **Network Security & Ethical Hacking:** Hashing algorithms, Wireshark, DoS/DdoS, SQL injection, Diffie-Hellman
 - **Data Analytics Tools:** AWS Quicksight, Tableau, Excel for Analytics
 - **Machine Learning:** scikit-learn, TensorFlow, PyTorch, OpenCV
 - **Programming:** Python, R, Java, SQL, C++
 - **Data Engineering:** Apache Spark, Kafka, Hadoop, Hive
 - **Web Tech:** Django, Flask, ReactJS
 - **Educational Skills:** Course Development, Outcome-Based Education (OBE), Project Supervision, Curriculum Renewal
 - **Instructional Delivery:** Online, Blended, In-person
 - **Team-based project supervision**
 - **Student collaboration tools** (e.g., GitHub classroom, Jira)
 - **Tool usage in class** (e.g., UML tools, IDEs, version control)

WORK EXPERIENCE | 10+ Years in HEC Recognized Universities/Colleges

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- Assistant Professor Computer Science at University of Lahore, Lahore Pakistan. August 2025 till date.
 - Lecturer (Undergraduate Students) Computer Science BPS-17 Regular at PHED-Govt. Rabia Basri Graduate College for Women Walton Road Lahore, Pakistan. July 2019 – July 2025.
 - Taught core and advanced courses: Artificial Intelligence, Big Data Analytics, Data Structures, Software Engineering, and Web Technologies
 - Designed and delivered outcome-aligned lectures using real-world tools like TensorFlow, Apache Spark, and AWS
 - Supervised capstone/FYP projects including: *VisionCraft* (AI-powered image generation), *SafeSpeak* (emergency voice recognition system), and *Bookify* (real-time inventory library system)
 - Integrated project-based learning and hybrid delivery methods to support diverse student needs
 - Lecturer (Undergraduate Students) Computer Science BPS-17 Regular at Govt. Degree College for Women Heir Badian Lahore, Pakistan. Sept 2017 till July 2019.
 - Visiting Lecturer (Undergraduate Students) Computer Science NCBA&E DHA Campus Lahore, Pakistan. Oct 2017 till Sep 2018.
 - Lecturer (Undergraduate Students) Computer Science BPS-17 Regular at Govt. P/G College for Women Bahawal Nagar, Pakistan. Dec 2015- Sept 2017.

PEER REVIEWER EXPERIENCE IN SCHOLARLY JOURNALS | 2019 till date

As an enthusiastic contributor to the academic community, I have actively engaged in voluntary work as a manuscript reviewer for various reputable journals, including IEEE Access, the International

Journal of Business Intelligence Research, and the Jordanian Journal of Computers and Information Technology. Having reviewed over 60 manuscripts, I have honed my skills in critically evaluating research submissions, providing constructive feedback, and ensuring the scholarly rigor and quality of published work. My dedication to maintaining the highest standards of academic rigor is reflected in my ongoing contributions.

TEACHING PORTFOLIO

- **Core Programming** (C++, Java, Python)
- **Web Technologies** (HTML5, CSS, JavaScript, PHP, React)
- **Advanced Software Development**
- **Big Data Analytics** (Apache Spark, MongoDB)
- **Data Structures & Algorithms**
- **Artificial Intelligence & NLP**
- **Software Engineering & SDLC**
- **FYP I & FYP II Supervision**
- Delivered lectures both **in-person and via digital platforms**, incorporating OBE and blended learning strategies.

SUPERVISED CAPSTONE PROJECTS

As part of my role as a Lecturer in the Department of Computer Science, I have supervised several high-impact final-year projects that integrate emerging technologies with practical applications to solve real-world problems:

- **Candy Crush game (C++ with graphics):** supervised/mentored as a software project with focus on state handling, modular design, and game logic.
- **VisionCraft App** - An Android-based AI application designed to transform user prompts into visuals using deep learning. Integrated features include text-to-image generation, face swapping (via *InsightFace*), and a personalized user dashboard. **Technologies:** TensorFlow, Python, OpenCV, Jupyter Notebook, Java, .NET Core 6, XML, MySQL, ReactJS.
Focus: Generative AI, Image Processing, User Experience Design
- **SafeSpeak App** - A women's safety Android app that automatically places emergency calls when a pre-configured keyword is detected through voice input.
Focus: Voice Activation, Android Security, Emergency Systems
- **Bookify – Library Management System** - A responsive, multi-device platform enabling real-time inventory management, automated borrowing/returning, search/discovery, analytics, and user tracking. **Technologies:** Java, HTML, CSS, JavaScript, MySQL.
Focus: Information Systems, Data Management, User Interaction
- **Community Care App** - A donation management Android app aimed at reducing food and clothing waste by connecting donors with those in need. Includes logistics support and location tracking for donation collection. **Technologies:** Android Studio, Java, XML, MySQL.
Focus: Social Good, Resource Optimization, Community Engagement
- **PakTutorHub Platform** - A web-based tutor-matching system allowing students to search, filter, and connect with tutors based on subject, location, budget, and availability. Implements advanced filtering and ranking algorithms for tutor profiles. **Technologies:** Node.js, Express.js, Tailwind CSS, JavaScript, HTML, MySQL.
Focus: Educational Technology, Web Development, Algorithm Design

COMMUNITY ENGAGEMENT & PROFESSIONAL SERVICE

- Reviewer for IEEE Access and IEEE Transactions on Knowledge and Data Engineering
- Remote mentor for international students from UCLA, University of Toronto, and George Mason University in AI, programming, and software engineering
- Engaged in freelance instruction for global learners, fostering inclusive digital learning spaces.